Alessandro Conti

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General Information

Birth Day	2002/01/11
Nationality	Italy
Languages	Italian, English

Summary

Master's degree in Computer Science and Engineering from the Politecnico di Milano, with a solid background in *cybersecurity*, *operating systems*, and *software engineering*. Passionate about low-level programming, system architecture, and the development of secure and scalable software. Motivated to apply the academic rigour and practical experience gained while writing my thesis to solve complex technical challenges in dynamic engineering contexts.

Work Experience

IT Consultant Aug 2026 - present
Spike Reply S.r.l – Reply group

- IT consulting projects in the IAM field
- API Security Governance Activities

Projects

CryptAttackTester Repository
This project is my master's thesis. The aim was to integrate parallelization for the existing ISD algorithms into the CryptAttackTester framework

Constant_Time-Parallel_Shuffling Repository
Shuffling arrays in constant time is a common problem in modern cryptography. This project involves analysing the technique proposed by Daniel J. Bernstein in Python, and implementing it in C employing parallelization.
The project includes porting the DJB code from Python to C, in both sequential and parallel versions. In addition to simple transcription, the project also analyses the computational complexities of the algorithms, providing a basis for optimisation and benchmarking.

Progetto_API_2023 Repository
This project manages a motorway with single-distance stations, each equipped with electric rental vehicles with variable range. A journey is a sequence of stages with no possibility of turning back, renting a vehicle at each stage. The goal is to find the route with the minimum number of stages between two stations; in the event of a tie, preference is given to the stages closest to the start of the motorway.

Progetto_Ingegneria_del_Software_2023 Repository

This project implements an online version of the board game 'My Shelfie'.

Progetto_Reti_Logiche_2023 Repository

This project implements a hardware module in VHDL that acts as an interface between a memory and four output channels. The module receives serial instructions relating to a memory address and the desired output channel.
The system receives input data via a single serial bit indicating the memory address from which to retrieve the data and the corresponding output channel. The system outputs consist of four parallel channels that transmit the entire memory word.

Sorting_Network Repository

This project implements several sorting networks.
The sorting networks implemented are: odd-even transposition sort, odd-even mergesort, bitonic sort, LS3 sort, 4-way mergesort, rotatesort, 3n sort of Schnorr and Shamir, 2D odd-even transposition sort and shearsort.

Certification

Kong Gateway Operations – Kong Inc. Aug 2026
Kong Gateway Foundations – Kong Inc. Aug 2026

Education

Computer Science and Engineering - Ingegneria Informatica (110/110) 2023 - 2026/07/22
Master of Science
Politecnico di Milano

Ingegneria Informatica 2020 - 2023/09/27
Bachelor's Degree
Politecnico di Milano

Liceo Scientifico opzione Scienze Applicate 2015 - 2020
High School
I.I.S. James Clerk Maxwell, Milan

Skills

Programming Language	C, C++, Java, Python, Bash, JavaScript
Frontend Languages	HTML, CSS
Query Languages	SQL
Operating System	Linux, Windows, macOS
Documentation	Latex, Markdown
Version Control	Git, GitHub
Database Management System	MySQL
Software	WireShark, TeamViewer, Microsoft Office

Last updated: 2026/09/05